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The Combined use of Measurements and Simulations for the Low-Uncertainty Characterization of a Reference Source of Electromagnetic Field

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***Abstract*—The accurate characterization of a broadband, compact and active antenna designed for use as a travelling sample for interlaboratory comparisons is described. Characterization is based on the combined use of measurement and simulations. The result of the characterization is the value and uncertainty of the generated electromagnetic field at different distances and in different electromagnetic environments (fully- and semi-anechoic). The dominant contribution to the uncertainty of the generated electromagnetic field is the mismatch between the output of the internal power source and the input of the radiating structure. Reduction of mismatch error cannot be easily achieved due to the physical limits imposed by the simultaneous requirements for a compact (small antenna size) and battery operated (low power source) travelling sample.**

Keywords—reference electromagnetic field, measurement uncertainty, simulation uncertainty, finite integral technique, method of moments, electromagnetic compatibility measurements.

I. Introduction

A compact, stable, active and broadband reference source of electromagnetic field in the frequency range from 50 MHz to 6 GHz was designed and realized for use as a travelling sample in interlaboratory comparisons (ILCs) [1, 2]. The travelling sample generates a known (with uncertainty) electromagnetic field that has been determined through the combined use of power, near- and far-field radiofrequency measurements and simulations performed by using two simulation tools employing entirely independent numerical methods. The result of this characterization is a calculable reference source of electromagnetic field.

The need for a calculable travelling sample for ILC application is justified by several reasons. The first reason is that a low-uncertainty characterization is required, where low means less than 2 dB, i.e. small with respect to the ordinary measurement uncertainty declared by the laboratories participating in the ILC (i.e. electromagnetic compatibility – EMC – test laboratories) which is in the range comprised between 5 dB and 6 dB. This 2 dB target uncertainty is not practically achievable by electromagnetic field measurements in ordinary test sites, especially for frequencies below 300 MHz, where the non-ideal behavior of anechoic chambers is the dominant contribution to measurement uncertainty and, by itself, accounts for 3 dB to 4 dB expanded uncertainty [3]. The second reason is that the electromagnetic field generated by the source must be calculable at the different distances (3 m and 10 m from the source) and in the different electromagnetic environments (fully-anechoic and semi-anechoic rooms) presently adopted in the standard EMC test activity. The third reason is that knowing in advance, rather than at the end of the ILC (i.e. several months after the start, through a statistical analysis of the measurement results provided by the participants), the electromagnetic field generated by the travelling sample permits to immediately verify if the measurement results provided by each participant are right or wrong and get confidence on the correct operation of the travelling sample after transportation. Further, the agreement between the calculated value of the electromagnetic field and the one obtained through the statistical analysis of the measurement results provides evidence to the participants about the competence of the coordinator of the ILC. As an example of this comparison see Table I where the difference between the average value and the calculated value is reported at the ten measurement frequencies selected for a recent ILC which started in August 2018 and concluded in February 2019 (24 participants).

TABLE I. Comparison Between the Average Measurement Result and the Reference Value

f MHz	δ^* dB	s^* dB	U dB	z' 1
50	-0.5	2.6	2.0	0.4
150	-0.6	1.5	2.0	0.6
250	-0.2	1.5	2.0	0.2
500	1.3	1.5	2.0	-1.2
750	0.2	1.7	2.0	-0.2
950	-0.4	1.4	2.0	0.4
2000	0.5	2.1	2.0	-0.4
3000	-0.2	1.6	2.0	0.1
5000	1.1	3.0	2.0	-0.9
6000	1.2	2.6	2.0	-0.9

The meaning of the quantities reported in Table I is as follows: f is the measurement frequency (in MHz), δ^* is the deviation between the average of the measured values (calculated by using robust statistics, as implemented through Algorithm A in Annex C of [4]) and the calculated value, s^* is the (robust, see again [4]) standard deviation of the measured values, U is the expanded uncertainty (corresponding to 95 % coverage probability) assigned to the calculated value and z' is a measure of relative deviation given by [4]

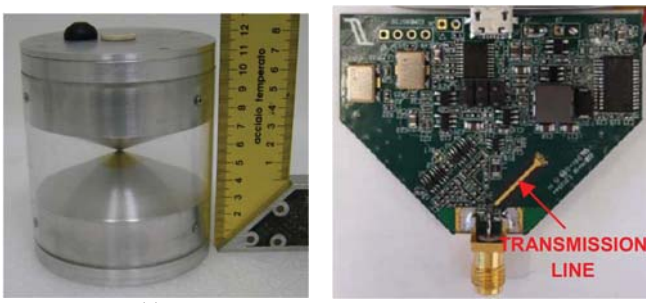
$$z' = \frac{\delta^*}{\sqrt{(U/2)^2 + \left(1.25\, s^*/\sqrt{p}\right)^2}}$$

where p is the number of participants ($p = 24$). The relative deviation is acceptable provided that $|z'| \le 2$, which is actually verified at each frequency.

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The scope of this contribution is to describe how measurements and simulations can be employed for the low-uncertainty characterization of an active calculable source of electromagnetic fields. An original aspect of the analysis is the evaluation of the uncertainty of a simulation which is based on a similar procedure to the one followed for the evaluation of the uncertainty of a measurement (i.e. the procedure described in the Guide to the expression of Uncertainty in Measurement, the GUM [5], and its Supplement 1, GUMS1 [6]). The travelling sample consists of the combination of a comb generator and a radiating structure (biconical antenna), see Fig. 1.



(a) (b)

Fig. 1. Biconical antenna (a) and comb generator (b). The comb generator feeds the biconical antenna and it is assembled inside the top cone.

The paper is organized as follows. Section II is devoted to the calibration of the comb generator. The prediction of the radiated field and the quantification of its uncertainty are reported in section III. In section IV a comparison between predictions and measurements is offered. Conclusion follows in section V.

II. Calibration of the Comb Generator

The comb generator was calibrated for the power delivered to a 50 Ω load at each of its harmonics (i.e. at frequencies multiple of 50 MHz, from 50 MHz to 6 GHz, 120 harmonics). Calibration was carried out comparing, frequency by frequency, by using a spectrum analyzer, the power of the harmonics with the output power of a calibrated sinusoidal signal generator. The output power of the signal generator was calibrated by using a high-accuracy power meter.

The calibration was performed in three steps and each step was repeated for each harmonic frequency. In the first step the power (in dBm) of each harmonic generated by the comb generator is measured by using the spectrum analyzer. In the second step the signal generator is driven to deliver to the input of the spectrum analyzer the same power value measured in the first step. The difference Δ (in dB) between the power displayed by the spectrum analyzer in the first and second steps is obtained. In the third step the output power of the signal generator (driven as in the second) is measured by using the power meter. If P_g is the output power of the signal generator then, to a first approximation, the power of the harmonic of the comb generator is $P_g + \Delta$. When measuring power with the spectrum analyzer the output of the comb generator, in the first step, and the output of the signal generator, in the second step, were connected to the spectrum analyzer input through the same cascaded connection consisting of (from the generator output to the spectrum analyzer input) a precision 10 dB attenuator, a low-loss coaxial cable and a DC-block. In the third step the power meter was connected to the signal generator directly through the 10 dB attenuator (no cable, no DC-block).

The scheme shown in Fig. 2 represents the connection between a source and a load through a generic two-port network. In the first step the source is the comb generator, the load is the spectrum analyzer and the two-port network is the cascade of the attenuator, cable and DC-block. In the second step the comb generator is replaced by the signal generator (the rest of the measurement chain being the same as in the first step). In the third step the source is the signal generator, the load is the power meter and the two-port network is the 10 dB attenuator. The general scheme in Fig. 2 applies to each of the three previously described measurement steps. From the same scheme a mathematical description of the power flow from a source to a load can be derived.

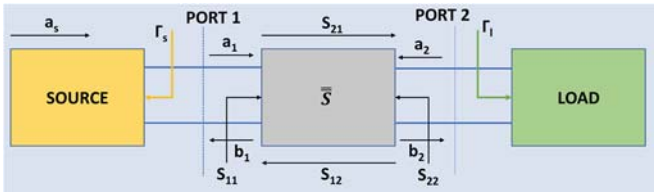


Fig. 2. Electrical connection scheme of a source to a load (measuring instrument) through a network characterized by S-parameters.

From the analysis of the scheme we indeed obtain (all terms are in logarithmic units)

$$P_s = P_l + A + M, \tag{1}$$

where $P_s = |a_s^2|$ is the power delivered by the source to a 50 Ω load, $P_l = |b_2^2|$ is the forward power to the load, $A = -20\log|S_{21}|$ is the attenuation of the two-port network and M is the mismatch (multiple reflection) error correction which is given by

$$M = 20log|(1 - S_{11}\Gamma_s)(1 - S_{22}\Gamma_l) - S_{12}S_{21}\Gamma_s\Gamma_l|. \tag{2}$$

By adapting (2) to each of the three measurement steps and combining the resulting equations to derive the comb generator power P_c we obtain

$$P_c = P_w + A_0 + \Delta + M_c, \tag{3}$$

where P_w is the power indicated by the power meter, A_0 is the attenuation of the 10 dB attenuator, Δ is the difference between the power indicated by the spectrum analyzer in the first and second steps and M_c is a term that accounts for mismatch correction in the first and second steps (M_{12}) and the third (M_3). We have

$$M_c = M_{12} + M_3, \tag{4}$$

where

$$M_{12} = 20\log\left|\frac{(1 - S'_{11}\Gamma_c)(1 - S'_{22}\Gamma_{sa}) - S'_{12}S'_{21}\Gamma_c\Gamma_{sa}}{(1 - S'_{11}\Gamma_g)(1 - S'_{22}\Gamma_{sa}) - S'_{12}S'_{21}\Gamma_g\Gamma_{sa}}\right| \tag{5}$$

and

$$M_3 = 20\log\left|(1 - S''_{11}\Gamma_g)(1 - S''_{22}\Gamma_w) - S''_{12}S''_{21}\Gamma_g\Gamma_w\right|. \tag{6}$$

$\Gamma_c, \Gamma_g, \Gamma_{sa}$ and Γ_w represent the output reflection coefficient of the comb generator and of the signal generator and the input reflection coefficient of the spectrum analyzer and the power meter, respectively. The S-parameters with a single quote correspond to the connection between the source and

the load in the first and second steps. The S-parameters with a double quote correspond to the 10 dB attenuator. It is worth noting that, assuming small mismatches,

$$M_{12} \cong 20\log\left|1 + S'_{11}\left(\Gamma_g - \Gamma_c\right) + (S'_{21})^2\Gamma_{sa}\left(\Gamma_g - \Gamma_c\right)\right|. \qquad (7)$$

We see from (7) that in order to minimize mismatch error in the first two measurement steps a good match at the source side ($|S'_{11}| \ll 1$) and a reasonably high insertion loss ($|S'_{21}| \ll 1$) are required. Mismatch error is relatively insensitive to the match at the load side ($|S'_{22}\Gamma_{sa}|$) since a cancellation occurs when taking the difference of the measurement results in the first and second steps. Further mismatch error is comprised between the limits

$$M_c|_{\min} = 20\log\left|(1-\delta)\frac{1-\frac{\alpha}{1-\gamma}}{1+\frac{\beta}{1-\gamma}}\right|,$$

$$M_c|_{\max} = 20\log\left|(1+\delta)\frac{1+\frac{\alpha}{1-\gamma}}{1-\frac{\beta}{1-\gamma}}\right| \qquad (8)$$

where $\alpha = |S'_{11}|_{\max}|\Gamma_c|_{\max} + |(S'_{21})^2|_{\max}|\Gamma_c|_{\max}|\Gamma_{sa}|_{\max}$, $\beta = |S'_{11}|_{\max}|\Gamma_g|_{\max} + |(S'_{21})^2|_{\max}|\Gamma_g|_{\max}|\Gamma_{sa}|_{\max}$, $\gamma = |S'_{22}|_{\max}|\Gamma_{sa}|_{\max}$, $\delta = |S''_{11}|_{\max}|\Gamma_g|_{\max} + |S''_{22}|_{\max}|\Gamma_w|_{\max} + |(S''_{21})^2|_{\max}|\Gamma_g|_{\max}|\Gamma_w|_{\max}$, and maxima are taken over the full frequency range of interest here (50 MHz to 6 GHz). The terms $|\Gamma_g|_{\max}$, $|\Gamma_{sa}|_{\max}$ and $|\Gamma_w|_{\max}$ were derived from the signal generator and the spectrum analyzer specifications, respectively, while the other terms were obtained from vector network analyzer (VNA) measurements. The limits (8) were found to be ±0.45 dB.

The measurement model equation is obtained from (3) including two corrections, namely the correction for the power meter inaccuracy δw and the correction for spectrum analyzer scale linearity error $\delta\Delta$. Therefore

$$P_c = P_w + A_0 + \Delta + M_c + \delta w + \delta\Delta. \qquad (9)$$

Calculation of measurement uncertainty was carried out by following the guidelines of the GUMS1 [6]. A probability density function (PDF) is assigned to each input quantity (the variables at the right side of (9)) according to the available information (measurement, calibration and specification data). Monte Carlo random number generation was then used to propagate the PDFs of the input quantities through (9) thus obtaining the PDF of the measurand P_c. Table II shows the assigned PDFs: shifted and scaled Student's *t* with four degrees of freedom (symbol t_4), rectangular within given limits (symbol *R*) and normal with given mean and standard deviation (symbol *N*). The best estimate of the comb generator output power is obtained adding the average of five readings of the power meter $\overline{P_{w_k}}$ with the average of five readings of the spectrum analyzer measured deviation $\overline{\Delta_k}$ ($k = 1,2,\ldots 5$). Note from Table II that the dominant contribution to measurement uncertainty is due to mismatch. The calculated standard uncertainty is 0.33 dB and the expanded uncertainty (corresponding to 95 % coverage probability) is 0.62 dB, worst case over all the 120 harmonics in the frequency range from 50 MHz to 6 GHz.

TABLE II. COMB GENERATOR POWER MEASUREMENT UNCERTAINTY BUDGET.

Quantity	Measurement unit	PDF	Mean	Standard uncertainty (dB)
P_w	dBm	$t_4\left(\overline{P_{w_k}}, \frac{0.18}{\sqrt{5}}\right)$	$\overline{P_{w_k}}$	0.11
A_0	dB	$R(10 \pm 0.1)$	10	0.06
Δ	dB	$t_4\left(\overline{\Delta_k}, \frac{0.16}{\sqrt{5}}\right)$	$\overline{\Delta_k}$	0.10
M_c	dB	$R(\pm 0.45)$	0	0.26
δw	dB	$N(0,0.05)$	0	0.05
$\delta\Delta$	dB	$R(\pm 0.2)$	0	0.12
P_c	dBm	$PDF\ (P_c)$	$\overline{P_c}$	0.33

III. PREDICTON OF THE ELECTROMAGNETIC FIELD RADIATED BY THE SAMPLE

A commercial electromagnetic simulation tool (CST® Studio Suite, Microwave Studio Module) was used, together with the result of the calibration of the comb generator, to predict the electric field radiated by the sample. The numerical algorithm adopted by the tool is based on the finite integral technique (FIT), which consists in the time domain solution of Maxwell integral equations over a finite size, discretized volume surrounding the radiator at discrete time steps. A comparison was carried out with a second electromagnetic simulation tool (FEKO®) based on the method of moments (MoM) in frequency domain, which is an entirely different numerical technique from the FIT adopted by CST® The scope is to evaluate the obtained degree of accuracy of the numerical predictions by comparing the results obtained using two independent methods.

Three electromagnetic configurations were analyzed for the radiated field, all of them in vertical polarization: 3 m distance in free-space (fully anechoic environment, from 50 MHz to 6 GHz), 3 m and 10 m distance, with the base of the sample at 80 cm above a perfectly reflecting and infinite ground plane (semi-anechoic environment, from 50 MHz to 1 GHz). Note that the maximum of the electric field in the height range between 1 m and 4 m was calculated in the semi-anechoic environment, as required by the applicable EMC standard [7].

A. *Fully anechoic environment*

A cubic volume of 1.5 m side and centered on the sample was discretized in about $43 \cdot 10^6$ hexahedral cells having a side comprised between 0.1 mm and 2 mm. Allowance for cell refinement around the corners of the sample was provided.

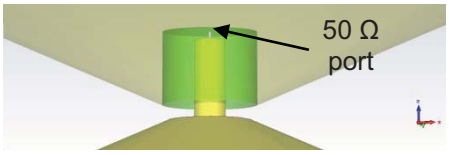


Fig. 3. Detail of the transition from the comb generator output reference plane to the biconical antenna input, as implemented in the computer design.

Appropriate boundary conditions were set at the borders of the volume to simulate free space. Electric and magnetic symmetry was employed to reduce memory occupation and computation time. The time step corresponding to the adopted resolution of the space discretization was 1.18 ps and the analyzed time interval was about 11 ns, i.e. well beyond the duration of the impulse response of the biconical antenna. The total computation time on a portable PC was about ninety minutes. All the relevant geometrical and physical (material) details of the biconical antenna were implemented in the computer design. The accurate description of the feeding of the biconical antenna was particularly critical due to its impact on the prediction of mismatch. The actual transition from the reference plane of the comb generator output to the input of the biconical antenna, consisting in a coaxial structure with rexolite as insulating material, was implemented in the computer design, see Fig. 3. The feeding is realized through a 50 Ω port connected between the inner coaxial conductor and the inner surface of the top cone (see Fig. 3). As a reference for dimensions in Fig. 3 note that the size of the gap between the tips of the cones is 0.55 mm. The simulation model equation is given by

$$E = E_{\mathrm{sim}} + P_c - 30 - M_a + \delta E + \delta P. \qquad (10)$$

where E is the predicted electric field (in dB(μV/m)), E_{sim} is the electric field that results from the simulation (in dB(μV/m)), P_c is the comb generator output power (in dBm), 30 (in dBm units) corresponds to the value of power set in simulations for feeding the biconical antenna (1 W), $M_a = 20\log|1 - \Gamma_c\Gamma_a|$ is the mismatch error correction (Γ_c is the measured output reflection coefficient of the comb generator and Γ_a is the reflection coefficient of the biconical antenna as seen from the 50 Ω port in Fig. 3), δE is a correction that takes into account the unavoidable deviation between the actual geometry of the biconical antenna and the one implemented in the computer design, δP is a correction associated to the deviation between the power generated by the comb generator during the calibration and when mounted inside the cone.

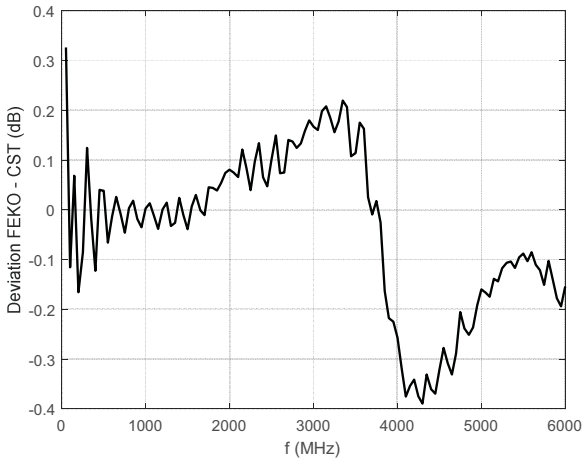


Fig. 4. Deviation between FEKO® and CST® as a function of frequency.

E_{sim} is affected by an error due to the numerical method on which the simulation tool is based and the chosen settings. This error was evaluated performing the simulation of the same computer design with another simulation tool based on MoM in frequency domain (i.e. an independent numerical method). The maximum deviation between the two simulations was less than 0.5 dB (see Fig. 4) over the full frequency range from 50 MHz to 6 GHz.

The maximum deviation associated with the correction δE was evaluated by performing a set of parametric simulations assuming different dimensions of the biconical antenna. It was realized that the parameter on which the simulated electric field is mostly sensitive is the gap between the tips of the cones, the one which is also less easily controllable when assembling the antenna. The interval of the possible values of the gap size was estimated to be within 0.4 mm and 0.7 mm and four simulations were carried out at 6 GHz assuming a gap size of 0.4, 0.5, 0.6 and 0.7 mm. The corresponding range of the values of the simulated electric field was less than 0.6 dB. Finally, the maximum deviation associated with the correction δP was estimated measuring the comb generator power in absence and in presence of a nearby grounded metallic structure. The metal enveloped the comb generator close to its terminals for the purpose of reproducing the effect of the coupling with the inner surface of the cone hosting the generator. The comb generator power was mostly sensitive to the presence of the surrounding metal in the frequency range between 5.2 GHz and 5.8 GHz. It is interesting to note that the shorted transmission line that is alternatively charged and discharged to generate the repeated impulses at the output of the comb generator, see Fig. 1(b), is one fourth of wavelength at about 5.5 GHz.

The uncertainty of the predicted electric field was quantified by using the Monte Carlo numerical technique along the guidelines offered by the GUMS1 and based on the simulation model (10). The corresponding uncertainty budget is shown in Table III. Note that the PDF of P_c is the one numerically obtained as described in the previous section II. $\overline{E_{\mathrm{sim}}}$ is the average value of the simulated electric field resulting from the parametric (at varying gap size) analysis. The dominant contribution to the uncertainty of the prediction is mismatch. The maximum of the product $|\Gamma_c\Gamma_a|$ over the full 50 MHz to 6 GHz frequency was evaluated to be 0.14. The corresponding limits of a U-shaped PDF are ±1.23 dB (see Fig. 5, where a plot of both $|\Gamma_a|$ and $|\Gamma_c|$ is shown).

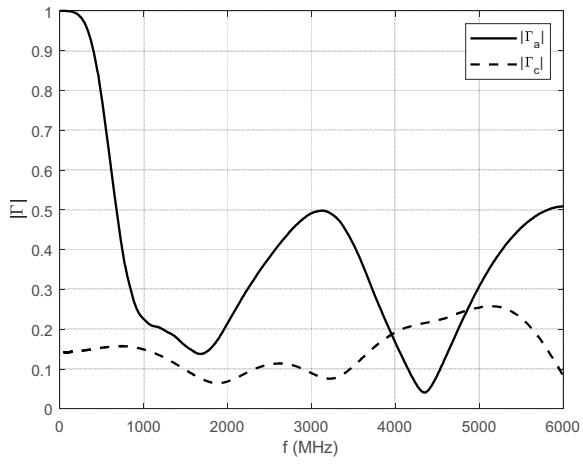


Fig. 5. Magnitude of the reflection coefficient of the antenna input and the comb generator output.

Finally, the standard uncertainty of the predicted electric field turns out to be 1.00 dB and the expanded uncertainty, corresponding to 95 % coverage probability, is 1.78 dB (worst case from 50 MHz to 6 GHz).

TABLE III. ELECTRIC FIELD PREDICTION UNCERTAINTY BUDGET.

Quantity	Measurement unit	PDF	Mean	Standard uncertainty (dB)
E_{sim}	dB(μV/m)	$R(\overline{E_{sim}} \pm 0.5)$	$\overline{E_{sim}}$	0.29
P_c	dB	$PDF\ (P_c)$	$\overline{P_c}$	0.33
30	dBm	---	30	0
M_a	dB	$U(\pm 1.23)$	0	0.87
δE	dB	$R(\pm 0.3)$	0	0.17
δP	dB	$R(\pm 0.3)$	0	0.17
E	dB(μV/m)	$PDF\ (E)$	$\bar{E}$	1.00

B. Semi-anechoic environment

We skip the details of the simulation settings for the semi-anechoic environment, which are similar to those used for the fully anechoic environment. It is instead worth mentioning the line of reasoning followed to get insight into the simulation results and assign an uncertainty figure to them. It is reasonable to assume, due to the physics of the electromagnetic problem, that the biconical antenna radiates in the semi-anechoic environment as in the fully anechoic one. More specifically the input impedance and radiation pattern of the antenna are not expected to significantly change when passing from an environment to the other. Further, the receiving antenna is in the far field of both the biconical antenna and its image. Then it is expected that the electric field generated by the biconical antenna in the semi-anechoic environment is obtained through scaling the electric field predicted in the fully anechoic environment by a factor that depends only on geometry and frequency and given by [8]

$$K_v = d_3 \left| \frac{e^{-j\beta r}}{r}\left(\frac{d}{r}\right)^2 + \frac{e^{-j\beta r'}}{r'}\left(\frac{d}{r'}\right)^2 \right|_{\max}, \qquad (11)$$

where $d_3 = 3$ m, d is 3 m or 10 m, r is the distance between the biconical antenna and the point where the field is measured, r' is the same distance but from the image of the biconical antenna, $\beta = 2\pi/\lambda$ and λ is the wavelength. (11) applies to the adopted vertical polarization and to a dipole like ($\sin\theta$) radiation pattern, which is indeed, to a good approximation, the shape of the pattern of the biconical antenna up to 1 GHz. The maximum is taken over the scanning height of the receiving antenna, i.e. 1 m to 4 m as required in [7]. It turns out that the ratio of the simulated fields in the semi-anechoic and fully anechoic environments deviates from (11) by no more than 0.40 dB (from 50 MHz to 1 GHz, for both the 3 m and 10 m distances). Hence, attaching this worst case 0.40 dB deviation to the uncertainty of the prediction in the semi-anechoic environment, only a slightly larger uncertainty results. This larger uncertainty is estimated as $\sqrt{\left(\frac{0.4}{\sqrt{3}}\right)^2 + 1^2} = 1.03$ dB in terms of one standard deviation, or 1.85 dB in terms of expanded uncertainty (corresponding to a coverage probability of 95 %).

IV. VALIDATION OF PREDICTION RESULTS

The electric field prediction results were validated through a comparison with measurement results. Measurements were carried out in a fully anechoic chamber at 3 m distance from the sample in the frequency range from 350 MHz to 6 GHz, where the imperfection of the chamber is tolerable. The prediction at the six frequencies from 50 MHz to 300 MHz was validated through comparison with near-field magnetic field measurements which were performed by using a highly symmetric loop antenna. These near field measurements provided an indirect confirmation of the validity of the prediction, assuming that the predictions have the same degree of accuracy both in the far-field and in the near-field. This is reasonable because the far-field predictions at locations outside the discretized volume are obtained from near-field (i.e. inside the volume) predictions.

We here briefly present the results of the measurements in the far field (i.e. in the fully anechoic chamber) and the comparison with predictions. Measurements were carried out by using three different types of antennas in order to cover the frequency range from 350 MHz to 6 GHz, namely a log-periodic dipole array from 350 MHz to 3 GHz (band #1) and two standard-gain horns, one from 3050 MHz to 4000 MHz (band #2) and another one from 4050 MHz to 6000 MHz (band #3). The reason behind the selection of these antennas is their relatively high gain, compared with other available (e.g., compact biconical) antennas. The advantages associated with the high gain are a relatively small sensitivity to anechoic chamber imperfections and a relatively high received power over the noise floor of the receiver (a good quality spectrum analyzer).

The measurement model equation is

$$E_m = V_{sa} + A + AF + M_m + \delta SA, \qquad (12)$$

where E_m is the measured electric field, V_{sa} is the voltage indicated by the spectrum analyzer, A is the attenuation of the cable connecting the receiving antenna to the spectrum analyzer, AF is the antenna factor of the receiving antenna, M_m is the mismatch error correction and δSA is the correction for spectrum analyzer inaccuracy. The measurement uncertainty budget is shown in Table III. Where three values are shown then the first applies to band #1, the second to band #2 and the third to band #3. When a single value is shown then it applies to the full frequency range.

TABLE IV. FULLY ANECHOIC CHAMBER ELECTRIC FIELD MEASUREMENT UNCERTAINTY BUDGET.

Quantity	Measurement unit	PDF	Mean	Standard uncertainty (dB)
V_{SA}	dB(μV)	$t_2\left(\overline{V_{SA}}, \frac{0.57}{\sqrt{3}}\right)$ $t_2\left(\overline{V_{SA}}, \frac{0.27}{\sqrt{3}}\right)$ $t_2\left(\overline{V_{SA}}, \frac{0.44}{\sqrt{3}}\right)$	$\overline{V_{SA}}$	0.33 0.16 0.25
A	dB	$N(\bar{A}, 0.06)$	$\bar{A}$	0.06

AF	dB(1/m)	$N(\overline{AF}, 0.35)$ $R(\overline{AF} \pm 0.3)$ $R(\overline{AF} \pm 0.3)$	$\overline{AF}$	0.35 0.17 0.17
M_m	dB	$R(\pm 1.0)$ $R(\pm 0.5)$ $R(\pm 0.5)$	0	0.58 0.29 0.29
δSA	dB	$N(0,0.23)$ $N(0,0.71)$ $N(0,0.71)$	0	0.23 0.71 0.71
E_m	dB(μV/m)	$PDF\ (E_m)$	$\bar{E}$	0.98 0.92 1.01

The measurement uncertainty due to the imperfection of the anechoic chamber was attached to the quantity V_{SA}. In order to evaluate this uncertainty contribution the receiving antenna was moved horizontally along the line passing through the center of phase of the receiving and the transmitting (travelling sample) antenna and vertically. The voltage received was then corrected for the far-field 1/distance decay and the deviation from the voltage received at the reference (3 m distance) position was evaluated. The displacements were 10 cm (along vertical) and 20 cm (along horizontal). A shifted and scaled Student's t PDF was assigned to this uncertainty contribution, as indicated in the second row of Table IV.

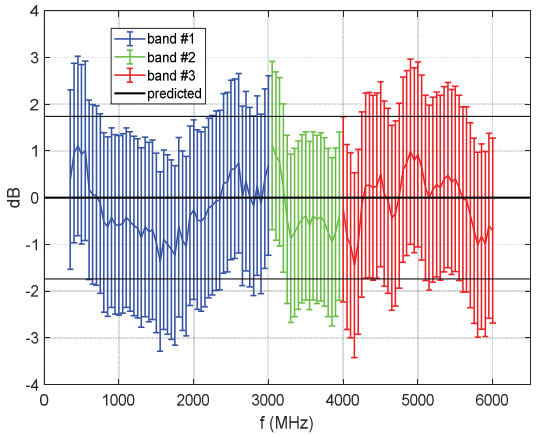


Fig. 6. Comparison between far-field measurements in a fully-anechoic chamber and the corresponding predictions.

The expanded uncertainty of the measured electric field obtained through application of the GUMS1 procedure and corresponding to a coverage probability of 95 % was 1.92 dB in band #1, 1.80 dB in band #2 and 1.98 dB in band #3. The result of the comparison between measured and predicted values of the electric field is shown in Fig. 6. Results are reported in term of deviation from the predicted values. Different colors correspond to different bands (as detailed in the legend). The length of the error bars is two-times the expanded measurement uncertainty. The range of the interval delimited by the black horizontal lines is two times the expanded prediction uncertainty. Full compatibility results thus providing confidence on the validity of both measurements and predictions.

V. Conclusion

The accurate characterization of a broadband source of electromagnetic field can be conveniently carried out by combining simulations and measurements. Modern electromagnetic simulation tools permit to predict the parameters of antennas with a high degree of accuracy provided that the simulated structure is known and simple enough to be fully represented by the computer design. Evaluation of prediction uncertainty follows the same established method of evaluation of measurement uncertainty, based on a model equation and propagation of uncertainty. The intrinsic error of the simulation can be evaluated by comparing the results produced by two independent (i.e. based on two entirely different numerical techniques) simulation tools. The dominant factor limiting the accuracy of the prediction is the mismatch between the source output and antenna input. This uncertainty contribution cannot be easily reduced. The requirement for a compact antenna implies a large mismatch especially at low-frequency, where the antenna size is small with respect to wavelength. Further, the reduction of the mismatch of the comb generator output can be achieved through a matching pad but at the expense of a loss of signal power (which is already relatively low since the source is battery operated).

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